

# JEFFREY TSENG

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Purdue University

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## PROFESSIONAL SUMMARY

Full stack developer with hands-on experience leading and developing user-first apps and websites, RAG backends, and defensive backend radar systems, seeking a 2026 software engineering internship.

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## EDUCATION

**Purdue University**, West Lafayette, IN Graduation: May 2029  
Bachelor of Science in Computer Science and Mathematics GPA: 4.0/4.0  
**Relevant Classes:** CS390: Web Applications Programming, CS193: Computer Science Tools, CS180: Problem Solving & Object-Oriented Programming, MA261: Multivariable Calculus

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## RELEVANT EXPERIENCE

**Hack the Future** Purdue University  
*Team Lead* October 2025 – Present

- **Led cross-functional development teams** of 10+ student engineers to design and deploy full-stack applications for local nonprofits, integrating **React, Node.js/FastAPI, and MongoDB**
- Conducted **code reviews** and managed teams with **Git workflow**, fostering a **collaborative development environment**
- Collaborated directly with nonprofit clients **to implement user-centric solutions** that fit their needs

**Google Developer Groups** Purdue University  
*Technical Teaching Assistant* August 2025 – Present

- Designed and deployed take-home projects for **300+ students teaching React, Google Cloud, Vertex AI, & Tensorflow**
- **Mentored 150+** developer role students, providing technical assistance on **Flutter, Firebase, and full-stack projects**
- **Selected as one of two technical mentors** from over 250+ applicants

**MIT Beaver Works** Cambridge, MA  
*Synthetic Aperture Radar Developer* July 2024 – August 2024

- Developed radar imaging **back-projection algorithms in Python and C++** using NumPy, SciPy, and Matlab to process radar data and generate images, processing radar data enabling **high-resolution images** of 2.6 in. soda cans across 10 by 10 meters
- Developed Synthetic Aperture Radar (SAR) applications **in collaboration with MIT Lincoln Labs** and the Department of Defense **to identify foreign objects through foliage**
- Built an F550 Hexacopter, with a **PulsOn 440 radar** with a **Raspberry Pi via socket programming** for **live data transmission**

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## PERSONAL PROJECTS

### UPlate

- Engineered an **AI-powered meal planner** by integrating the **Google Gemini Generative AI API** to generate **personalized daily meal plans** tailored to user-calculated macronutrient goals
- **Developed from the ground up using Flutter** to implement a **custom, responsive UI** featuring a stateful, multi-page onboarding flow to seamlessly **collect user biometrics**
- Implemented an offline-first data architecture **using SQLite** for instant app responsiveness and persistence, **with asynchronous data synchronization** to a **Firestore cloud backend**

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- Architected and deployed a dynamic, **full-stack personal portfolio website** to showcase skills and professional experience
- Engineered the frontend with **React and TypeScript**, creating a responsive, mobile-first user interface
- Developed a **robust backend using Node.js** to handle API requests and serve project data, ensuring efficient performance

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## TECHNICAL SKILLS

**Languages:** Python, Java, HTML, CSS, C++, C, TypeScript

**Libraries and Tools:** NumPy, SciPy, Matlab, Gazebo, ROS, Node, React, Tailwind

References Available Upon Request